

IGCSE Exam: Answer 2

- (d) When eight bytes of data have been collected, they are transmitted to a computer 100km away. Parity checks are carried out to identify if the data has been transmitted correctly. The system uses **even parity** and column 1 is the parity bit.

The eight bytes of data are sent together with a ninth parity byte:

	parity bit	column 2	column 3	column 4	column 5	column 6	column 7	column 8
byte 1	1	0	0	0	0	1	0	0
byte 2	1	1	1	1	0	0	1	1
byte 3	0	1	0	0	1	0	0	0
byte 4	0	1	1	1	0	0	0	1
byte 5	1	0	0	0	1	1	1	1
byte 6	0	0	0	0	0	0	0	0
byte 7	1	1	1	0	1	0	0	0
byte 8	1	0	0	0	1	1	1	0
parity byte	1	0	1	1	0	1	1	1

- (i) Identify which of the eight bytes contains an error.

byte (byte) 5 [1]

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43

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- (ii) Identify which column contains an error.

column (column) 4 [1]

- (iii) The incorrect bit is indicated where the byte number and column cross.

Give the corrected byte.

[1 0 0 1 1 1 1 1] [1]

- (iv) Calculate the denary value of the corrected byte.

..... 1 5 9 [1]

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44

IGCSE Exam: Answer 4

Question	Answer	Marks
8(a)	– Odd – Odd – Even – Even	4
8(b)	Any one from: – there is a transposition of bits – it does not check the order of the bits (just the sum of 1s/0s) – even number of bits change – incorrect bits still add up to correct parity	1
8(c)(i)	Four from: – Multiple bits are sent at the same time – Uses multiple wires – Data is sent in both directions ... – ... but only one direction at a time	4
8(c)(ii)	Any two from: – Bits may arrive skewed – More expensive to setup/manufacture/purchase cable – Limited distance – More prone to interference/error	2

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45

IGCSE Exam: Answer 5

Question	Answer	Marks
6(a)	– Odd – Even – Even – Odd	4
6(b)	Any one from: – There is a transposition of bits – Bits still add up to correct parity	1
6(c)(i)	– Data is sent one bit at a time – Data is sent using a single wire – Data is sent in both direction ... – ... at the same time	4
6(c)(ii)	Any one from: – Data transmission can be slower (than parallel) – Additional data may need to be sent	1

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46

IGCSE Exam: Answer 6

Question	Answer	Marks
9	- 1 - 0 - 1 - 1	4

IGCSE Exam: Answer 7

Question	Answer	Marks
9	1 mark per each correct parity bit: Parity bit Register A 1 0 1 0 0 1 0 1 Register B 1 1 0 0 0 0 0 1 Register C 1 1 0 0 0 0 1 1	3

IGCSE Exam: Answer 8

Question	Answer	Marks
5	- B Three from: - Added up the number of 1's / 0's in each register - With the parity bit, two registers have an odd number of 1's / 0's - One register has an even number of 1's / 0's - Odd parity must be the parity used	4

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49

Automatic Repeat Request (ARQ): IGCSE Exam: Answer 10

Question	Answer	Marks																																
2(a)	2 marks for 3 correct bits, 1 mark for 2 correct bits <table><tr><td>Parity Bit</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table>	Parity Bit								0	1	0	1	0	0	1	1	0	1	0	1	1	1	1	1	1	1	0	1	0	0	0	1	2
Parity Bit																																		
0	1	0	1	0	0	1	1																											
0	1	0	1	1	1	1	1																											
1	1	0	1	0	0	0	1																											
2(b)	Two from: ☐ Set of rules for controlling error checking/detection // it's an error detection method // used to detect errors ☐ Uses acknowledgement and timeout ☐ Request is sent (with data) requiring acknowledgement ☐ If no response/acknowledgment within certain time frame data package is resent ☐ When data received contains an error a request is sent (automatically) to resend the data ☐ The resend request is repeatedly sent until packet is received error free/limit is reached/acknowledgement received	2																																
2(c)	Checksum	1																																

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50

2.2 Methods of error detection

2.2.3 Checksum

IGCSE Past Paper Exam Answer 11

PUBLISHED		
Question	Answer	Marks
7(a)	- Data is sent multiple bits at a time ... down multiple wires - Data is sent in one direction only	3
7(b)	Any two from: - Parallel is not suitable for long distances // only suitable for short distances ... the data could become skewed - Parallel cables are not manufactured above approx. 5m	2
7(c)	Any three from (MAX 2 for ARQ): - Checksum used to detect errors (during transmission) ... using a calculated value - ARQ checks if data is received ... uses acknowledgement and timeout ... requests data be sent again if (checksum) detects error / not received	3

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51

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Answer 12

- 1 (a) 1 mark for the correct working in BOTH parts
 1 mark for valid
 1 mark for not valid

Identification number 1: working

$$= (4 \times 6) + (2 \times 5) + (1 \times 4) + (9 \times 3) + (2 \times 2) + (3 \times 1)$$

$$= 24 + 10 + 4 + 27 + 4 + 3$$

$$= 72 \div 11$$

$$= 6 \text{ remainder } 6$$
valid/not valid: NOT valid

Identification number 2: working

$$= (8 \times 6) + (2 \times 5) + (0 \times 4) + (1 \times 3) + (5 \times 2) + (6 \times 1)$$

$$= 48 + 10 + 0 + 3 + 10 + 6$$

$$= 77 \div 11$$

$$= 7 \text{ remainder } 0$$
valid/not valid: VALID

[3]

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52

Parity/Check digit/Check Sum and ARQ:IGCSE Exam: Answer 12

(b) 1 mark for correct working + 1 mark for check digit

working

$$= (5 \times 6) + (0 \times 5) + (2 \times 4) + (4 \times 3) + (1 \times 2)$$

$$= 30 + 0 + 8 + 12 + 2$$

$$= 52$$

need to add 3 to make the total 55 (i.e. exactly divisible by 11)

check digit: 3

[2]

(c) 1 mark for each description and example

2 digits transposed

(e.g. 280419 becomes 280149/two digits have been switched)

incorrect digit

(e.g. 280419 becomes 250419/one of the digits has been mistyped)

[2]

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53

IGCSE Exam: Answer 16

Question	Answer	Marks
9(a)(i)	Two from: ☐ Data is transmitted one bit at a time ☐ Data is transmitted using a single wire ☐ Bits arrive in order/sequence	2
9(a)(ii)	Two from: ☐ Data is transmitted multiple bits at a time/simultaneously ☐ Data is transmitted using multiple wires ☐ Bits may arrive out of sequence/skewed (and are reordered)	2
9(a)(iii)	1 mark for each: ☐ Data is transmitted in both directions ☐ ... at the same time/simultaneously	2

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IGCSE Exam: Answer 16

Question	Answer	Marks
9(b)	Maximum of three marks per error detection method. 1 mark for naming the method, 2 marks for describing it. Parity (check) ☐ Odd or even parity can be used ☐ Bits are added together // 1 bits are counted ☐ Parity bit added (depending on parity set) ☐ Parity checked on receipt ☐ If parity bit is incorrect an error is detected Checksum ☐ Calculation performed on data (to get the checksum) ☐ Checksum sent with data ☐ Checksum recalculated after transmission ☐ Comparison made between checksum before and checksum after transmission ☐ Error detected if checksums are different Automatic repeat request (ARQ) ☐ Uses acknowledgement and timeout ☐ Request is sent (with data) requiring acknowledgement ☐ If no response/acknowledgment within certain time frame data package is resent ☐ When data received contains an error a request is sent (automatically) to resend the data ☐ The resend request is repeatedly sent until packet is received error free/limit is reached/acknowledgement received	9

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55

IGCSE Exam: Answer 17

Question	Answer	Marks
5	1 mark for correct register, 3 marks for reason: - Register Y Any three from: - Count the number of 1/0 bits (in each byte/register) - Two bytes/registers have an odd number of 1/0 bits // Two have odd parity - Even parity must be the parity used - One byte/register has an even number of 1/0 bits // One uses even parity - The two with an odd number of one bits/odd parity are incorrect // Register X and Z should have even parity	4

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56

IGCSE Exam: Answer 18

Question	Answer	Marks															
4(a)(i)	<table border="1"> <thead> <tr> <th>Received Byte</th><th>Transmitted correctly (✓)</th><th>Transmitted incorrectly (✓)</th></tr> </thead> <tbody> <tr> <td>10001011</td><td></td><td>✓</td></tr> <tr> <td>10101110</td><td>✓</td><td></td></tr> <tr> <td>01011101</td><td>✓</td><td></td></tr> <tr> <td>00100101</td><td>✓</td><td></td></tr> </tbody> </table>	Received Byte	Transmitted correctly (✓)	Transmitted incorrectly (✓)	10001011		✓	10101110	✓		01011101	✓		00100101	✓		4
Received Byte	Transmitted correctly (✓)	Transmitted incorrectly (✓)															
10001011		✓															
10101110	✓																
01011101	✓																
00100101	✓																
4(a)(ii)	One from: - ARQ - Check Sum	1															

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57

IGCSE Exam: Answer 19

Question	Answer	Marks												
5(a)	1 mark per correct tick <table border="1"> <thead> <tr> <th>Received byte</th><th>corrupted during transmission (✓)</th><th>not corrupted during transmission (✓)</th></tr> </thead> <tbody> <tr> <td>10110100</td><td>✓</td><td></td></tr> <tr> <td>01101101</td><td></td><td>✓</td></tr> <tr> <td>10000001</td><td>✓</td><td></td></tr> </tbody> </table>	Received byte	corrupted during transmission (✓)	not corrupted during transmission (✓)	10110100	✓		01101101		✓	10000001	✓		3
Received byte	corrupted during transmission (✓)	not corrupted during transmission (✓)												
10110100	✓													
01101101		✓												
10000001	✓													
5(b)	Four from: ☐ Uses acknowledgement and time out ☐ Check performed on received data // error is detected by e.g. parity check, check sum ☐ If error detected, request sent to resend data // negative acknowledgment is used ☐ If no acknowledgement is sent that data is received // positive acknowledgment is used ☐ Data is resent / Resend request repeated, till data is resent correctly ... ☐ ... or request times out // limit is reached	4												

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58

IGCSE Exam: Answer 20

- 9 (a) 1 mark for correct check digit and 1 mark for showing the calculation

$$\begin{aligned}
 &(4 \times 1) + (2 \times 2) + (4 \times 3) + (1 \times 4) + (5 \times 5) + (0 \times 6) + (8 \times 7) \\
 &= 4 + 4 + 12 + 4 + 25 + 0 + 56 = 105 \\
 &105/11 = 9 \text{ remainder } 6 \\
 &\text{check digit is: } 6
 \end{aligned}$$

1 mark for any correct line of working

[2]

- (b) incorrect check digit

[1]

- check digit should be 1
- $(3 \times 1) + (2 \times 2) + (4 \times 3) + (0 \times 4) + (0 \times 5) + (4 \times 6) + (5 \times 7) // 3 + 4 + 12 + 0 + 0 + 24 + 35 //$
Total = 78
- 78/11 gives 7 remainder 1

[2]

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59

IGCSE Exam: Answer 21

- 4 (a) Intersection of Row 7 and column 4 circled

[1]

- (b) – Row (byte number) 7 has an odd number of 1s (five 1s)
– Column (bit number) 4 has an odd number of 1s (five 1s)

[2]

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60

IGCSE Exam: Answer 22

9 (a)

Binary number A:

1	1	1	0	0	1	0
---	---	---	---	---	---	---

Binary number B:

1	0	0	1	1	1	0
---	---	---	---	---	---	---

[2]

(b)

Parity Bit

Binary number A

1

Binary number B

1

[2]

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61

IGCSE Exam: Answer 23

5 (a) 1 mark per correctly placed tick

Received byte	Byte transmitted correctly	Byte transmitted incorrectly
1 1 0 0 1 0 0 0		✓
0 1 1 1 1 1 0 0		✓
0 1 1 0 1 0 0 1	✓	

[3]

(b) (i) byte number: 7

column number: 6

[2]

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62

IGCSE Exam: Answer 23

(ii) Any **two** from:

- letter "A"(byte 7) transmitted as odd parity (three 1s)
- column 6 has odd parity (seven 1s)
- intersection of byte 7 and column 6 indicates incorrect bit value

[2]

(c) 190

[1]

(d) Any **one** from:

- 2 bits interchanged (e.g. $1 \rightarrow 0$ and $0 \rightarrow 1$) that won't change parity value
- even number of bits/digits are transposed
- If there are multiple errors in the same byte/column, that still produce the same parity bit, the error will not be detected

[1]

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63

IGCSE Exam: Answer 24

12 (a)

1	1	1	1	1	0	0	0
0	0	0	0	0	1	1	1

[2]

(b) 1 mark for error detection method and 1 mark for description

- Check sum
- ... sum of bits is transmitted and checked against the sum of the received bits
- Check digit
- ... a digit that is calculated (e.g. using modulo-11) and transmitted with the data
- ARQ
- ... when an error is detected in a packet of data a request is automatically sent for the data to be resent

[2]

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64

IGCSE Exam: Answer 25

7 (a) (i) 1 mark for correct check digit and 1 mark for showing the calculation

$$(4 \times 1) + (2 \times 2) + (4 \times 3) + (1 \times 4) + (5 \times 5) + (0 \times 6) + (8 \times 7)$$

$$= 4 + 4 + 12 + 4 + 25 + 0 + 56 = 105$$

$$105/11 = 9 \text{ remainder } 6$$

check digit is: 6

[2]

(ii) 1 mark

– No/incorrect check digit

2 marks

– Total is 78

– $78/11 \dots$

– $\dots$ gives 7 remainder 1

– check digit should be 1

[3]

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65

IGCSE Exam: Answer 25

(b) (i) 1 mark for each correct parity bit

parity bit

0	1	1	0	0	1	1	0
---	---	---	---	---	---	---	---

parity bit

1	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---

[2]

(ii) Any **one** from:

- an even number of digits are changed
- a transposition error(s) has occurred

[1]

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66